MD-DGFLS: MULTI-DOMAIN DEEPFAKE FORENSIC VERIFICATION CERTIFICATE

IEEE TPAMI / CVPR Technical Forensics Specification v2.2-PROD | Cryptographic Audit Dossier

File Name:	da8068a5.jpg	**Target Domain:**	Face
Audit Timestamp:	2026-09-23 12:00:23 UTC	**SHA-256 Hash:**	c8c6ffedf6ddf55a47f57562...
Forensic Verdict:	**VERIFIED AUTHENTIC MEDIA**	**Confidence:**	**96.5%**
Tamper Type:	Authentic Human Subject	**Bounding Box:**	None

Three-Stage Visual Forensics Pipeline



Stage 1: Source Image



Stage 2: Tamper Overlay



Stage 3: XAI Heatmap

Mathematical Innovations & Multi-Expert Telemetry

CDFAM (Cross-Domain Alignment)	Contrastive Score: 0.9742 (Margin: m=0.3)
ARCC (Entropy Calibration)	Dynamic Calibration Temperature: T = 3.1099
TGHFL (Transformer-GAN Cross-Attention)	Cross-Attention Energy: 0.259
MEFCN (Multi-Expert Consensus)	{'Face': 96.5, 'Medical Radiograph': 1.9, 'Satellite Terrain': 1.4, 'Regular Out-of-Domain': 2.5, 'Natural Scene': 1.9}
HLRN (Localization Refinement)	Coarse-to-Fine Refinement: 1/1 Native
Frequency Stream (ELA & Laplacian)	ELA Discrepancy: 1.082 | Sensor Noise Var: 140.3

This certificate is cryptographically anchored and generated by the MD-DGFLS Autonomous Forensics Engine. Verified under IEEE TPAMI standards.